

Product Requirements Document

Friendly Friend

Shell Hacks

Team <Div> A

Project Manager: John Ruiz

Date of Creation: 09/24/2026

Version 1.0

Introduction

A mobile app inspired by small-town provinces in the Philippines to allow users to create their own personal villages with close friends to trade food available in their fridge to create the feeling of “one big fridge”. This app aims to promote community and lessen the financial burden of modern-day bulk grocery shopping for college students by offering an interactive UI that breaks the awkward barrier that may come from asking peers for help.

Problem Alignment

Problem

Most grocery stores pack items in bulk for the typical suburban family and cost more when individually packed, leaving college students shopping for one to overspend on groceries when sometimes they only need one more *egg* to finish the recipe.

Why this Matters

Going to the grocery store is not always an available option for college students who face financial burden and sometimes lack the transportation to even make it there in the first place. The bulk packaging can be unforgiving leaving students to overspend on something they may only need a little of. For those who can't make it to the grocery store, that one missing ingredient can be the line between having dinner that night or surviving until the next trip to the store.

Target User & Pain Points

Target Group: College Students from ages 18 – 25. Their personalities are friendly, and tired. Young adults want easier access to groceries and ingredients in those moments where only a few ingredients are missing.

Primary User: Young adults who have a solid network of friends and paying for their own groceries.

User Pain Points

- "I don't have eggs; I can't make that meal"
- "I ran out of instant noodles; I'll wait until Friday after class to buy more"
- "I forgot to buy flour, but the next bus comes during class."

Current Alternatives & Shortcomings

Online Grocery Shopping: While these tools allow users to buy groceries at any time at the click of a button, the delivery adds an extra cost and doesn't tackle the bulk packaging of most grocery products.

Texting Friends: While this service fulfills a similar function as the application, it leaves the user to take the first step into asking for help which can be awkward for many people of this age.

Solution Alignment

Product Vision

A mobile app inspired by small-town provinces in the Philippines in which users can add villagers to their village and trade food with one other similar to how children nowadays trade in the Roblox game adopt-me or fidget trading in person to create the feeling of "one big fridge" to promote community.

High-Level Solution

Users will download an app in which they create an account with a username, password, and phone number. After creating the account, they will go to a landing page with a button to add a villager, to access their own profile stats, and to access their own personal fridge. The village itself will always take up the main screen in which it will be a top-down view that users can interact with and zoom in and out and look at their island. They can add a villager to their village by using a unique friend code and in their personal fridge, they can add ingredients they have available for other users to see. They can click on a villager's hut to access their fridge and request any food they have available and any offerings they may have. Requests can be accepted, rejected, or countered by the fridge owner — see Trade Request Flow for the full logic. To add reliability, each profile will have user stats in which other users can see raw data of “thank you” received, requests completed, and photo gallery if a user chooses to take a photo once a request is completed.

Trade Request Flow (State Machine)

States : pending, countered, accepted, rejected, completed

Transitions:

Pending -> accepted

Pending -> rejected

Pending -> countered

Countered -> accepted / rejected (by receiver)

Countered -> countered (re-entered by either: receiver counters, or requester adds to their ask)

Accepted -> completed (requester confirms delivery – one sided confirmation)

Rules :

- Only the requester can mark a request “completed”
- Like + Photo Upload are only available once state = completed
- Rejected and completed are terminal – no reopening
- The requester cannot counter a counter, they can only add more to their request or reject the offer to counter
- The receiver can always counter once a request is made or is increased by a previous counter

Reputation / Profile Stats (No Score)

Profile displays raw stats only – no computed reputation score :

- Deals completed (count)
- Thank you's received (count)
- Photo Gallery (grid of uploaded completion photos if the user chooses to do so)

Rules:

- Thank You button only appears once a request reaches state = completed and can be used once per a request
- Only the requester can send a Thank You (mirrors one sided- complete confirmation)
- Thank You Count is additive only, cannot be removed once given
- Photo Gallery Pulls from photos uploaded during completed requests (via the Trade Request Flow)
- No score, ranking, or algorithmic trust rating in MVP

Scope

MVP Scope

- Sign Up & Login
 - o Username
 - o Password
 - o Phone Number
- Add villager Button
 - o Unique Friend Code
 - o Once added, can add the villagers to multiple villages
- Personal Fridge Button
 - o Add Item
 - Multiple Basic Options + Other
 - Add Title
 - o Item shows up on shelf
 - o On existing icons, increase or decrease count
 - When it reaches 0
 - Confirm to delete it
- Village Screen
 - o Ability to click on villager's house
 - Villager's Fridge
 - Request Food by clicking on icon
 - Request status and completion handled per Trade Request Flow.
- Profile Stats
 - o Display raw counts: request completed, thank you's received, photo gallery, and no computed score
- A way to organize villagers into separate villages.

- Label the villages
 - For people with multiple friend groups
- Trade requests and profile stats follow the logic defined in "Trade Request Flow" and "Profile Stats" sections above/below — not re-described here.

Future Scope

- Decorate your fridge

Out Of Scope

- Shared villages (life360-style shared visibility)

Tech Stack (Mobile App)

Front End:

- React Native + Expo

Back End:

- Node Js
- Tiger Data Postgre SQL (For a Hackathon Challenge)